

OSINT Collector App — Product Spec v0.1

1. Product Summary

The app is a project-based OSINT collection workspace powered by Chrome extension modules.

As the user browses sites like Google Maps, Yelp, Facebook, LinkedIn, business websites, registries, and other public sources, the extension activates the correct collector module for the current page. Each module gathers useful data from that page and sends observations into the main app.

The main app stores the data, connects entities into a graph, surfaces cross-project relationships, and helps the user query, explore, review, and export useful information.

The core idea is:

```
Modules collect facts.  
AI suggests meaning.  
The user confirms important interpretations.  
The graph preserves facts, relationships, and evidence.
```

2. Core Product Principle

The app should not feel like a collection of unrelated scrapers.

It should feel like one cohesive OSINT workspace where each module contributes to a larger intelligence graph.

```
Collector Module  
→ Observation  
→ Entity  
→ Relationship  
→ Evidence  
→ Project Layer  
→ Global Graph  
→ Useful Output
```

Each collector module is independent, but all modules speak the same shared data language.

3. Primary User Experience

The user works inside a project.

A project may represent:

- Client research
- Personal research
- Lead gathering
- Market mapping
- Decision-maker discovery
- Target investigation
- A business, person, domain, niche, or location being explored

A project is flexible. It is not limited to one type of workflow.

Project creation should be lightweight:

```
Project name  
+ optional goal  
+ optional starting target
```

Example:

```
Project: Davis County Contractors  
Goal: Find decision makers  
Starting target: concrete contractors in Davis County
```

4. Project-Based Graph Model

The system should have a global graph underneath and project-specific layers on top.

A node should not be duplicated just because it appears in more than one project.

Example global nodes:

```
Business: ABC Landscaping  
Website: abclandscaping.com  
Phone: 801-555-1234  
Person: John Smith
```

The same node may appear in multiple projects:

```
Project: Davis County Contractors
Project: Bad Website Leads
Project: Contractor Outreach June
```

The app should actively surface cross-project connections.

Examples:

```
This phone number appears in 3 projects.

This domain was already captured under another business.

This person is connected to two companies you researched separately.

This address links two businesses that looked unrelated.
```

Projects are not silos. They are focused layers over the larger intelligence graph.

5. Chrome Extension Shape

The Chrome extension should behave like a live side-panel workspace.

The side panel should stay open while browsing and should update based on the current tab.

The side panel should show:

```
Current project
Project goal
Active module
Current page/site context
Useful findings
Best Next Paths
Captured observations
Evidence capture tools
Navigation back to workspace/graph
```

The side panel is not just a small popup. It is the live working layer.

6. Module System

Modules are site-specific or context-specific.

Examples:

```
Google Maps → Google Maps Module
Yelp → Yelp Module
Facebook → Facebook Module
LinkedIn → LinkedIn Module
Known business website → Website Module
Unknown website → Generic Website Collector
```

Each module has its own behavior because each site requires different collection logic.

Example:

Google Maps may need:

```
Keyword list
Results table
Scroll & Collect
Capture visible results
Map/list result parsing
```

A generic website may need:

```
Current page scan
Useful link classification
Contact info extraction
Social link detection
Schema data detection
Best Next Paths
```

The shared side-panel shell stays consistent, but the module workspace changes based on the current site.

7. Collection Behavior

Collection behavior is module-specific.

Some modules can auto-capture lightweight data immediately.

Other modules may require an explicit user action.

Example:

Google Maps Module

- May require Scroll & Collect
- May collect based on selected keyword
- May show current results in a table

Website Module

- Can scan current page automatically
- Can classify useful links
- Can extract contact/social/schema data
- Can suggest Best Next Paths

The system should avoid forcing the user to manually save each item.

The better model is:

Observe automatically.
Store observations.
Surface what matters.
Let the user pin, ignore, review, or explore.

“Save” is not the main action. The app is already collecting.

Better user actions:

Pin
Ignore
Review
Explore
Open
Graph
Confirm
Reject

8. AI Handling and Trust Model

Anything AI creates should be listed as a suggestion, not treated as confirmed fact.

Confirmed facts include:

Visible page text
URLs
Phone numbers
Email addresses
Addresses
Page titles
Structured data
Direct source links
Direct scraped listing data

AI suggestions include:

Possible owner
Likely decision maker
Suggested relationship
Best Next Path ranking
Link classification
Lead/opportunity interpretation
Screenshot extraction
Summaries
Inferred business relationships

Graph relationships should support statuses:

confirmed
suggested
reviewed
rejected

Example:

ABC Landscaping → HAS_WEBSITE → abclandscaping.com
Status: confirmed
Evidence: Google Maps listing + website URL

John Smith → POSSIBLE_OWNER_OF → ABC Landscaping
Status: suggested
Evidence: About page + LinkedIn profile
Confidence: medium

The user should be able to confirm or reject AI-generated suggestions.

9. Evidence Model

The app should preserve where data came from.

Every important fact or relationship should have evidence attached.

Evidence may come from:

```
Live page capture
Screenshot paste
Copied text
Copied table
Manual note
Imported CSV
Page URL
HTML snapshot
Source timestamp
```

The extension should support an “Add Evidence” area in the side panel.

Evidence capture options:

```
Paste Screenshot
Paste Text
Paste Table
Add Note
```

AI can analyze pasted evidence and extract observations, but those AI-created observations should be treated as suggestions until confirmed.

Example:

```
User pastes screenshot of a Facebook page.
```

```
AI extracts:
```

- business name
- phone
- visible profile URL
- visible category
- possible person name
- visible post/date

The screenshot is stored as evidence.
The extracted facts become observations.
The inferred meaning becomes suggestions.

10. Website Module Concept

When the user is on a website, the module should focus on what is visible or discoverable from that site.

It should not show the full previous path, such as:

Google Maps → ABC Landscaping → Website

That path can be preserved in the background, but the module view should focus on:

Current site
Current page
Useful paths available from this page
Captured data from this page

Website Module Example

Project: Davis County Contractors
Goal: Find decision makers

Active Module: Website Collector
Site: abclandscaping.com
Page: Home
Status: Scanning useful paths

Page Map

The page tree should not show every raw link.

It should show classified useful paths.

Example:

abclandscaping.com
├─ Current Page ✓ scanned
| ├─ Phone ✓ found

		Address	✓ found
		Email	○ not found
		Schema Data	✓ found
		About / Company Page	○ available
		Contact Page	○ available
		Team / Staff Page	○ available
		LinkedIn Profile	○ available
		Facebook Page	○ available
		YouTube Channel	○ available

Legend:

- ✓ captured
- available

11. Best Next Paths

The app should not show raw link dumps.

Instead, AI should classify and rank links into "Best Next Paths."

Raw links:

84 page links found

AI-reduced output:

Best Next Paths

1. About / Company
Reason: likely identity, owner, history, or company background.
2. Team / Staff
Reason: likely people connected to the business.
3. Contact / Quote
Reason: likely direct contact method.
4. LinkedIn
Reason: likely company/person relationship data.

5. Facebook

Reason: likely social proof, activity, contact info.

Hidden unless expanded:

Privacy policy
Terms
Image links
Duplicate nav links
Tracking links
Blog tags
Footer junk

The app should collect broadly but display goal-relevant and useful paths first.

Clicking a Best Next Path should open it in a new browser tab.

Behavior:

Click Best Next Path
→ opens in new tab
→ extension detects new page
→ matching module activates
→ project context stays active
→ relationship path is stored

Example:

Website Module
Best Next Path: LinkedIn Company Page

Click Explore
→ opens LinkedIn in new tab
→ LinkedIn Module activates
→ project remains Davis County Contractors
→ graph records website → linked profile relationship

12. Relationship Trails

The graph should preserve discovery trails, not only isolated nodes.

Example trail:

```
Google Maps listing
→ business website
→ about page / footer
→ LinkedIn / Facebook / Instagram
→ person name
→ owner / founder / manager role
→ history / previous company / related entity
→ another project connection
```

This allows the user to ask:

```
Why do we think this person is connected?
Where did this relationship come from?
What page proved it?
What other projects has this appeared in?
```

The graph should support branches.

Example:

```
Business website
├─ About page → owner name
├─ Footer → phone/email/address
├─ Social links → Facebook/LinkedIn
├─ Blog/history → old company names
└─ Schema data → official business details
```

The side panel should show a small useful tree, but the full graph belongs in the main workspace.

13. Main Workspace

The main app should not be graph-only.

It should support:

```
Query/search collected data
Explore graph visually
Review captured data
```

Follow paths/branches
Build/export lead lists or reports

Core modes:

Project Workspace
Graph View
Data / Entities View
Suggestions View
Report / Export View

The most important long-term combination is:

Graph + query + useful output

14. Main Workspace Views

Project Workspace

Purpose:

Show the current project, goal, recent activity, important findings, suggested paths, and useful outputs.

Should include:

Project summary
Goal
Starting target
Recent observations
Useful connections
Pinned entities
Suggested AI findings
Cross-project matches
Open paths to explore

Graph View

Purpose:

Show entities and relationships visually.

Should include:

- Nodes
- Edges
- Relationship labels
- Confirmed vs suggested status
- Evidence indicators
- Project-layer filtering
- Cross-project relationship indicators

Data / Entities View

Purpose:

Search, filter, and inspect the collected data in a table/list format.

Entity types:

- Businesses
- People
- Websites
- Emails
- Phones
- Addresses
- Social Profiles
- Listings
- Reviews
- Keywords
- Pages
- Evidence

Suggestions View

Purpose:

Review AI-created suggestions before treating them as trusted.

Suggestion examples:

Possible owner
Possible duplicate
Possible same business
Suggested relationship
Useful next path
Possible decision maker
Cross-project match

Report / Export View

Purpose:

Turn collected data into something useful.

Potential outputs:

Lead list
Contact list
Research report
CSV export
Project summary
Graph screenshot/export
Evidence-backed findings

15. Initial MVP Scope

The first version should focus on proving the full system with a small number of modules.

Recommended MVP:

1. Project system
2. Chrome side panel shell
3. Google Maps Module
4. Generic Website Module
5. Observation API
6. Basic entity matching
7. Basic project graph
8. Best Next Paths
9. AI suggestions layer
10. Evidence paste capture

The first complete workflow:

Create project:
Davis County Contractors

Goal:
Find decision makers

Use Google Maps Module:
Search and collect businesses by keyword

Open a business website:
Website Module scans page

Website Module finds:
About page
Contact page
LinkedIn
Facebook
Phone
Address
Schema data

AI suggests:
Best Next Paths
Possible decision-maker paths
Possible cross-project matches

Main app shows:
Collected entities
Graph relationships
Suggestions
Exportable lead/research list

16. First Modules

Google Maps Module

Purpose:

Collect local business results from Google Maps.

Likely features:

Keyword list
Selected keyword focus
Visible results capture
Scroll & Collect
Business name
Category
Rating
Review count
Phone
Website
Address
Maps URL
Result position
Search location

Generic Website Module

Purpose:

Collect useful website data and suggest meaningful next paths.

Likely features:

Current page scan
Phone extraction
Email extraction
Address extraction
Schema.org extraction
Social link detection
Useful link classification
Best Next Paths
Contact page detection
About/team/company page detection
Tech stack detection later

Evidence Capture Module

Purpose:

Allow user to paste screenshots, text, tables, or notes and convert them into observations.

Likely features:

Paste screenshot
Paste text
Paste table
Add note
AI extraction
Attach evidence to project
Create suggested observations
Create suggested relationships

17. Data Model Concepts

Core records:

Project
Goal
Observation
Entity
Relationship
Evidence
Source
Suggestion
Module
Collector Run

Main entity types:

Business
Person
Website
Domain
Email
Phone
Address
Social Profile
Listing
Review
Keyword
Search Result
Page
Evidence Item

Relationship examples:

BUSINESS_HAS_WEBSITE
BUSINESS_HAS_PHONE
BUSINESS_HAS_EMAIL
BUSINESS_AT_ADDRESS
WEBSITE_LINKS_TO_SOCIAL
WEBSITE_HAS_PAGE
PAGE_MENTIONS_PERSON
PERSON_WORKS_AT_BUSINESS
PERSON_POSSIBLE_OWNER_OF_BUSINESS
BUSINESS_HAS_LISTING
ENTITY_APPEARS_IN_PROJECT
ENTITY_MATCHES_ENTITY

Relationship statuses:

confirmed
suggested
reviewed
rejected

18. Open Product Questions

These still need to be decided.

Naming

Current working labels:

Workspace
Graph
Best Next Paths
Active Module
Evidence
Suggestions

Need better final product language later.

Side Panel Navigation

Current idea:

New Project
Workspace
Graph

But wording may need refinement.

Graph Interface

Need to explore graph libraries and interface patterns.

Possible concerns:

Graph should not become chaotic
Must be easier to use than Kuzu-style graph explorer
Needs filtering, pinning, path trails, and evidence inspection

AI Suggestion Review

Need to decide how suggestions are reviewed:

Inline confirm/reject
Separate suggestions queue
Confidence badges
Evidence cards

Raw vs Filtered Data

The UI should show useful classified data first, but still allow access to raw collected data when needed.

19. Design Rules

1. The project context should always stay visible.
 2. The active module should always be clear.
 3. Modules should collect facts broadly.
 4. AI should interpret, rank, and suggest.
 5. AI-created meaning should not be treated as confirmed fact.
 6. The graph should preserve evidence and discovery trails.
 7. Cross-project connections are a core value.
 8. The side panel should help the user keep moving.
 9. The main workspace should support graph, query, review, and output.
 10. The app should avoid dumping raw scraped data unless the user asks for it.
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20. Product Definition

This app is a project-based OSINT collection and graph workspace.

It uses Chrome extension modules to gather public data while browsing, stores facts as observations, connects entities into a global graph, overlays project-specific layers, surfaces cross-project connections, and uses AI to suggest meaningful paths, relationships, and outputs while keeping AI-generated claims reviewable.